Paleomagnetic Results from the Lower Permian Dunkard Series of West Virginia

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Abstract. Measurements have been made of the direction of remanent magnetization in more than 100 field samples of Permian age from the Dunkard series of West Virginia in the eastern United States. Most samples have a reversed polarity (i.e., north-seeking vector pointing southward) but a few of the samples have a normal polarity, the first to be reported in Lower Permian rocks. Subsequent ac demagnetization experiments have shown the magnetization of most of these samples to be stable and suggest that some of the initial observed scatter in direction was due to small isothermal moments acquired by the samples before measurement. The resulting virtual magnetic pole is in good agreement with the mean pole derived from the published data from rocks in the western United States but is in disagreement with the other data from eastern North America. To resolve these differences the following seem to be necessary: (1) additional data from eastern North America and/or the Arctic; (2) a study of the NRM of associated Permian red beds and lavas; and (3) a study of the cause of the scatter present in the measurements from the southwestern United States.

Introduction. Up to the present time most of the Permian data for North America have come from the southwestern United States, largely from studies of the Abo and Supai formations [see *Graham,* 1955; *Doell,* 1955; *Runcorn,* 1955; *Collinson and Runcorn,* 1960; *Irving,* 1956, 1959]. These investigations indicate a north magnetic pole in Asia in the vicinity of 40°N, 110°E. The virtual pole positions found in these investigations scatter more than 30°, a scatter much too large to be attributed to sampling or instrumental errors. Some of this scatter may be due to variability of the field direction during the roughly 30 million years of the Permian, although this type of variability should have been accounted for in the initial means since large stratigraphic intervals were sampled. Thus it seems strange (particularly in view of the results given below) that most of the ovals of confidence do not touch, and this leads to uncertainty as to the significance of an over-all mean pole position.

One way to see if this much variability or scatter should be expected for Permian rocks and to check the dipolar nature of the field is to study the remanent magnetization of rocks of the same age from other widely separated localities. *Fuller* [1963] has cited some of Black's unpublished results from Prince Edward Island, Canada, and concludes that there may be some evidence that Permian rocks from Prince Edward Island and Greenland give results that are not in agreement with the data from the southwestern United States.

Although the Prince Edward Island locality is more suitable for such comparison by being father away from the southwestern United States than the localities in West Virginia chosen for the present study (4000 km versus 2500 km), the West Virginia strata are flat lying and undisturbed by later geologic events and thus are not subject to errors involving dip correction or tectonic rotation.

Summary of Dunkard stratigraphy. The stratigraphy of the sediments exposed in the Ohio River valley in Ohio and West Virginia has recently been studied by *Cross and Schemel* [1956], and their work has been relied upon for age assignment as well as location of sections suitable for paleomagnetic study. The summary below, and the generalized stratigraphic section of the lower half of the Dunkard series, shown in Figure 1, is largely condensed from their report.

The youngest strata, other than unconsolidated sands and gravel, in the Ohio River valley in Ohio and West Virginia belong to the Lower Permian Dunkard series. The general outcrop area (Figure 2) consists of about 20,000 km^2 in West Virginia, Ohio, and Pennsylvania. The

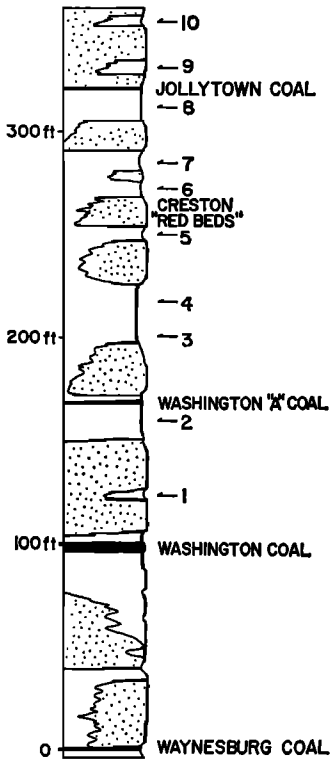


Fig. 1. Generalized stratigraphic section showing approximated stratigraphic position of samples studied.

thickness of Dunkard strata reaches a maximum of slightly less than 400 meters and apparently represents an essentially continuous period of terrestrial and fresh water deposition extending from latest Pennsylvanian time well into lower Permian time. The base of the Dunkard series is placed by definition at the top of the Waynesburg coal, which is about 30 meters below an occurrence of the plant *Callipteris conferta,* which is believed by most paleobotanists to be restricted to the Permian. No marine fossils have been reported from Dunkard rocks.

Four distinct sediment types are present in the Dunkard series cyclothems. A typical cyclothem is as follows (from bottom to top): (1) buff to white sandstone; (2) gray calcareous shale which may be locally red and may have intercalated fresh water limestone; (3) gray to green clayey siltstone and claystone; and (4) coal and overlying dark carbonaceous noncalcareous shale and siltstone. This is an idealized cyclothem, for many are in reality incomplete and have very rapid facies changes along strike. The red claystone weathers very rapidly to form a soft receding slope with little or no rock exposure. Thus one cannot find the red siltstone and claystone merely by locating the top of a sandstone but must look at many exposures before a section suitable for sampling is found. *Graham* [1955] reported measurements from a suite of samples of green to drab sandstones from the Dunkard series and found them to be greatly scattered and apparently unstable. Thus in the present study the drab sandstones associated with the red units were not sampled.

Description of sample localities. In the course of field sampling, more than thirty localities were visited. From these thirty localities five sampling areas were chosen that appeared to have enough material in the red shale facies to warrant further study. Three of these sampling areas are 1 to 2 miles east of Ben's Run, West Virginia, and are indicated in Figure 2 as sites 1, 2, and 3. The second sampling area is south of Parkersburg along West Virginia Route 2 about 5 miles south of Lubeck (site 4) and the third area is along West Virginia Route 20 about 5 miles southeast of New Martinsville, West Virginia (sites 5, 6, 7, 8, 9, and 10). The approximate stratigraphic position of the five sampling sites is shown by the arrows in Figure 1.

At each of these sampling sites four to twenty 2.54-cm-diameter samples 4 to 20 cm long were collected by means of a portable core drill. Con-

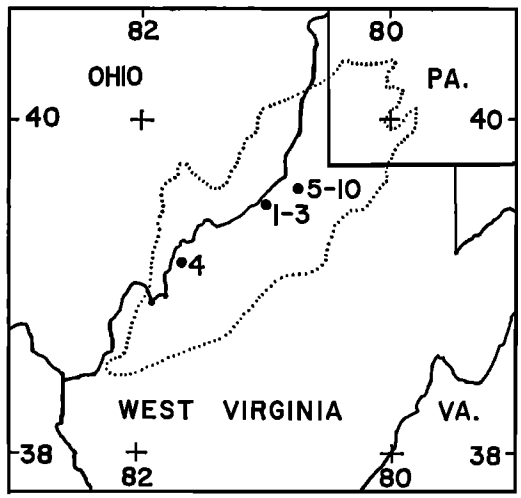


Fig. 2. Location map showing sample sites and area of outcrops, Dunkard series.

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siderable difficulty was experienced in sampling because these red claystones and siltstones disintegrate rapidly when exposed to water. It was found that careful regulation of the coolant water flow made it possible to sample almost any lithology if sufficient patience was used. However, it was much easier and faster to select only those lithologies that were at least slightly cemented, as indicated by a homogeneous, unfractured, and somewhat rounded weathering surface. During subsequent sample preparation, it was found that the use of a nonpolar solvent, such as kerosene, instead of water greatly reduced the amount of sample disintegration. The samples were oriented by means of a brass mark (or scratch) of known orientation along the length of the core before the sample was broken free from the enclosing rock.

Each field core was cut into one or more cylinders 2.54 cm high. Since these subsamples had a common orientation mark scratched on their sides before their removal in the field, no transfer of marks was required on about half of the samples. The other samples either had incomplete marks or were so fragile that it was necessary to encapsulate them in an epoxy plastic. In these cases new orientation marks were necessary; however, it is improbable that errors in excess of 2° were introduced, and, since it is unlikely that these errors were systematic, they should affect only the precision and not the accuracy of the measurements.

The NRM of the samples was measured with an air turbine spinner magnetometer of design similar to *Graham's* [1955] [*Helsley*, 1964]. Total angular error involved in all the marking and measuring operations is probably well within 3° for most of the samples.

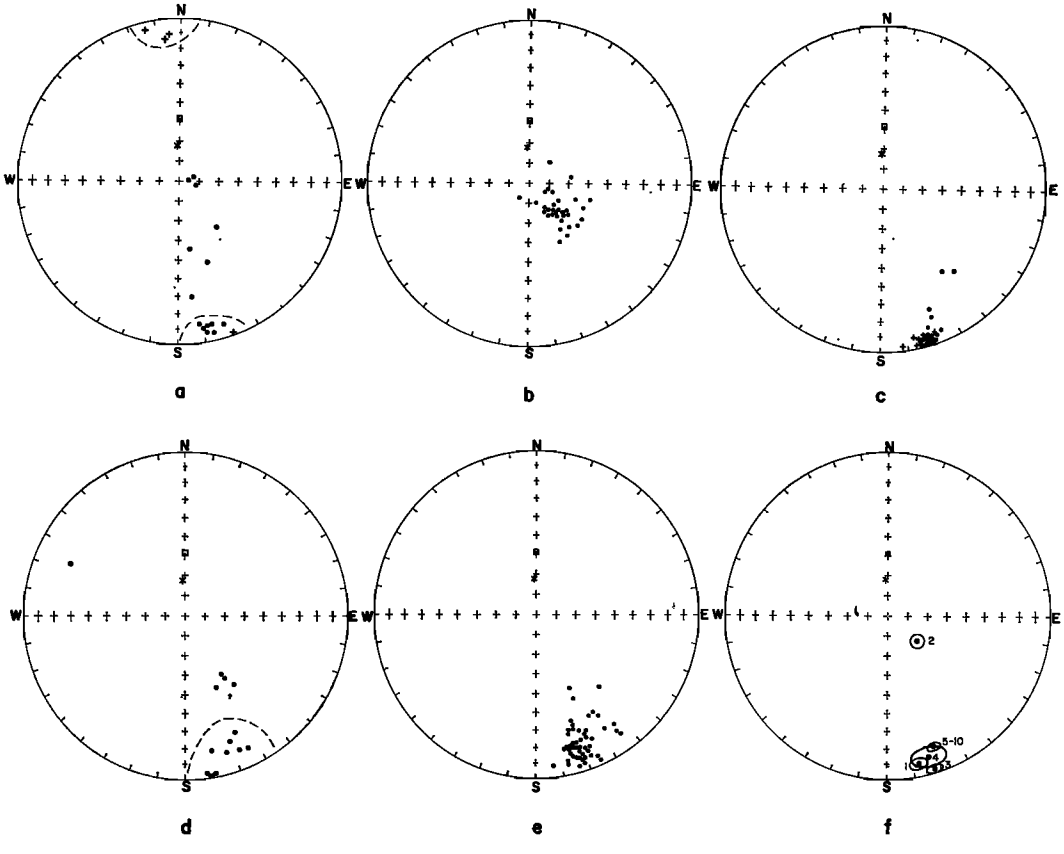


Fig. 3. Natural remanent magnetizations observed in rocks from the Dunkard series: (*a*) site 1, (*b*) site 2, (*c*) site 3, (*d*) site 4, (*e*) sites 5–10, (*f*) mean values. □ geocentric dipole, * present earth's field, + upper hemisphere, ● lower hemisphere. (These symbols are the same for all the figures except no. 9.)

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Results of NRM measurements. As is readily seen in the results shown in Figure 3, some instability or 'streaking' is present at two of the sites, and only those samples clustering within about 30° of the edge of the net (enclosed within dashed line) have been used in the calculation of the means of these sites. The other sites show no significant 'streaking' and thus might be expected to give a more reliable mean direction of magnetization. This is not the case, however; one of the sites has a steep inclination and the other have nearly horizontal inclinations. Further study of site 2 (Figure 3*b*) thus seemed advisable. At site 2 an 80-cm red claystone is exposed in a roadcut on the highway east of Ben's Run, West Virginia. It is both overlain and underlain by gray to gray-green claystone and siltstone, the lower unit being sandy about 1.5 m below the red claystone. The first set of samples taken from one outcrop at site 2 provided about half of the points shown in Figure 3*b*. The second group of samples came from other outcrops 3 to 30 meters west of the first outcrop and provided the remainder of the points shown in Figure 3*b*. Since the measurements from the first outcrop gave a relatively tight cluster of points, well removed from the normal Permian field direction, they were postulated to have acquired their magnetization through remagnetization by lightning. However, this does not seem to be the case, since other samples from the other outcrops of the same horizon gave identical results. Ten of these samples were subjected to ac demagnetization (see below) and were found to be unstable. It is surprising that these samples should be so tightly clustered about a position intermediate between the Permian and present-day field directions, for it seems unlikely that all samples could acquire approximately equal amounts of Permian and present-day field components. Since the mechanism by which these unstable samples became remagnetized remains unknown, these measurements have not been considered in the subsequent discussions regarding the position of the Permian virtual geomagnetic pole.

A second unexpected feature of the NRM measurements is shown in Figure 3*a*. Here, in addition to a weak cluster at the southeast edge of the net and an instability streak from there to the present field, three other points are present at an azimuth of 350°, lying upon the upper hemisphere. Their orientation is almost exactly 180° from the weak cluster at the southeast edge and thus represents a direction of magnetization of reversed polarity with respect to the main cluster. Since the main cluster at the southeast edge of the net is classed as being reversely magnetized, these three samples on the upper hemisphere near the northern edge of the net are normally magnetized. So far as is known, these are the first normally magnetized rocks of Permian age to be found on the North American continent and the only normally magnetized rocks of Lower Permian age anywhere in the world. Unfortunately, only three samples could be obtained from the red claystones at this stratigraphic horizon and no samples of the surrounding green claystones and siltstones were taken. Therefore, these three samples of normal polarity should be taken as preliminary results until samples from the adjacent green and gray sediments are measured.

The results shown in Figure 3 (*c* and *e*) are tightly clustered and show little 'streaking' and thus are assumed to represent the direction of stable magnetization acquired during Lower Permian time. Part of the samples shown in Figure 3*e* along with most of those shown in Figure 3*d* have been subjected to ac demagnetization as dicussed below; they show a small decrease in scatter and a change in mean direction of magnetization of about 8°, thus suggesting that they are indeed stably magnetized for the most part. No rocks suitable for field stability tests or tests for time of magnetization were found in the course of the field sampling. However, the presence of the normal and reversed samples in Figure 3*a* provides a powerful test of the magnetic stability of some of these rocks and provides some justification for the discarding of the 'streaked' directions. Nevertheless, it was impossible to test the assumption that the magnetization is of Lower Permian age.

The mean directions of magnetization before demagnetization were computed using the methods described by *Fisher* [1953, p. 303, 2nd equation] and are shown in Figure 3*f*. All of these means fall in a cluster at the southern edge of the net, and although all of their cones of confidence do not overlap there is probably no geophysically significant difference, for each of the localities is slightly different in age, as indicated in Figure 1. The means are skewed more

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in inclination (20°) than in declination (10°), and although this difference may not be significant it can be interpreted as the result of a small amount of instability being present in some of the samples, particularly in those of sites 5 to 10 (Figure 3), where all samples were used in the calculation of the mean. That this is indeed the case is shown by the results of the demagnetization experiments given below.

It should be noted that the precisions indicated may be unduly high (cones of confidence unduly small) as a result of treating each measured subsample as an independent measurement. Any other form of treatment, such as the use of field sample means, would have been equally arbitrary, for there is as much scatter between two samples from the same core as there is between adjacent or far-removed samples from the same horizon. The length of time represented by sedimentation at any one sampling site is necessarily short owing to the restricted development of the red claystone, and thus the site means may not represent an adequate sampling in time. The total length of magnetization time is unknown, for magnetization may have continued throughout much of the rock's diagenetic history.

The results in Figure 3*e* come from eight individual red claystone horizons separated stratigraphically by up to 25 meters. These eight horizons seem to be associated with only six cyclothems and thus, for statistical purposes, these results have been treated as six horizons. Two of these are from single thick claystone beds and the other four are from two or more adjacent thin beds. (See Table 1 for computed values.) The individual sample measurements show more scatter than the other localities, partly as the result of a small amount of instability being present, yet they do not have a sufficient scatter to include all of the measure-

TABLE 1. Source and Pole Data

Site	Lat.	Long.	Decl., deg	Incl., deg	α	k	N	Pole Position Lat. (N)	Pole Position Long. (E)	dm	dp
Before Demagnetization											
1_n*	39.5	279.0	175.0	9.7	9.30	176.9	3	45.6	101.9	9.4	4.7
1_r†	39.5	279.0	169.4	8.0	4.12	181.59	8	45.9	110.1	4.2	2.1
$1_{(n+r)}$	39.5	279.0	173.9	8.4	3.6	163.9	11	45.9	107.9	3.6	1.8
3	39.5	279.0	162.8	2.1	3.0	100.8	24	46.5	125.0		
4	39.2	278.3	163.8	11.3	6.7	60.3	9	42.5	117.5		
5–6	39.6	279.3	166.9	14.4	4.1	114.5	12				
7	39.6	279.3	161.0	21.5	5.4	65.9	12	39.0	121.0		
8	39.6	279.3	164.0	16.1	5.5	58.0	13				
9–10	39.6	279.3	160.1	13.6	6.5	63.9	9				
Mean of all sites											
	39.5	279.0	163.8	12.5	5.02	145.6	7	42.0	121.5	5.11	2.53
After Demagnetization											
4	39.2	278.3	165.5	1.0	5.10	173.4	6	48.0	121.5	5.1	2.6
5	39.6	279.3	161.9	10.3	15.81	34.8	4	42.2	124.0	16.0	8.1
6	39.6	279.3	159.3	9.8	14.05	318.2	2	41.5	127.4	14.2	7.2
7	39.6	279.3	164.0	14.3	16.56	229.4	2	40.9	120.5	17.0	8.7
8	39.6	279.3	157.5	3.5	19.86	160.2	2	43.8	131.3	19.9	10.0
9	39.6	279.3	163.5	5.7	8.05	131.2	4	44.9	122.9	8.1	4.1
10	39.6	279.3	168.9	13.8	34.58	54.3	2	42.3	114.3	35.3	18.1
Mean of demagnetized results											
	39.5	279.0	163.0	8.4	4.75	162.3	7	43.6	122.8	4.8	2.4
Mean of demagnetized results plus undemagnetized results from sites 1 and 3											
	39.5	279.0	163.5	7.7	3.88	176.8	9	44.1	122.3	3.9	2.0

* n = normal.
† r = reversed.

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ments from the other localities; e.g., there are no points on the upper hemisphere as there are at sites 1 and 3. Thus it is thought that most of the differences in the means should be attributed to small differences in time of magnetization and/or small amounts of instability. As is shown below, the instability alone is probably insufficient to account for all of the observed differences.

Demagnetization experiments. About one-fourth of the samples measured in the course of this study were subjected to ac demagnetization at the Socony Mobil Laboratories, Dallas, Texas, using the methods and equipment described by *Patton and Fitch* [1962]. The method is basically as follows: The sample is first measured for its NRM direction and intensity and is then placed in the demagnetizing coil in a magnetically shielded room where the dc field has been nulled to a few gammas. A predetermined amount of 400-cps ac current is then passed through the coil windings and allowed to decay slowly at a uniform rate. This produces a slowly and uniformly decaying ac field having a maximum intensity of up to 1400 oe. The sample is then removed from the coil and shielded room and its remaining NRM measured.

In these experiments the sample was not rotated during the demagnetization. On many samples, particularly at the higher field values, a second run was made at the same field intensity after the sample was rotated through 180°. No significant differences, i.e., differences of greater than 5°, were observed except at the fields of over 700 oe, and even then the differences were generally less than 10°. This implies that the demagnetization procedure was satisfactory; i.e., it was not introducing an anhysteretic component in the specimen.

Complete demagnetization runs for four samples are shown in Figure 4. Most of the change in direction and intensity occurred at ac field values of less than 350 oe. Beyond this point a continued slow decrease in intensity was noted but very little additional change in direction occurred. This is the behavior that would be expected if a 'soft' or isothermal component or components had been removed and only the 'hard' and presumably stable component remained. The change in direction with progressively higher fields indicates that a component in the general direction of the earth's present field was removed. It also suggests that another randomly oriented 'soft' component may have been present, for not all of the change lies within the great circle joining the present field direction with the cluster of points at the southeast edge of the net. This additional component is thought to have been acquired by the samples during storage in the laboratory before measurements were made.

The ability of these samples to acquire a small amount of isothermal magnetization is shown in Figures 5 and 6. The demagnetization curve of sample 46*B* is shown by the solid line in Figure 5. When the demagnetization was completed, the sample was placed in a known orientation in the earth's field and left overnight. A considerable amount of ac equipment was operating in the room and the power mains for the building were only a few meters away. As a result, the background ac field in the room was higher than for most laboratory areas, possibly by as much as a factor of 50. When the sample was remeasured the next morning, it was found to have an orientation given by the point '0' on the dotted curve. Subsequent demagnetization moved the magnetic moment toward the previous afternoon's value along a great-circle route. The intensity of the total isothermal component was about 1¾ times that of the remaining remanent component of the previous evening but only about 15% of the initial NRM of the sample. The field value of 175 oe which was necessary to remove the isothermal component was higher than would have been expected for a strictly isothermal magnetization of such a short time constant. Therefore, an anhysteretic component may also have been involved.

Many samples were measured twice during this study, and most of those that had high intensities (greater than 5×10^{-5} emu/cc) or belonged to the stable southeast cluster showed little or no change in direction. However, many of the samples chosen for the demagnetization experiments showed some instability (i.e., were members of the streaked or scattered population), and these generally did not produce the same results upon remeasurement. This can easily be seen in Figure 6, where the original measurements (open circles) are compared with the measurements (dots) made after the sam-

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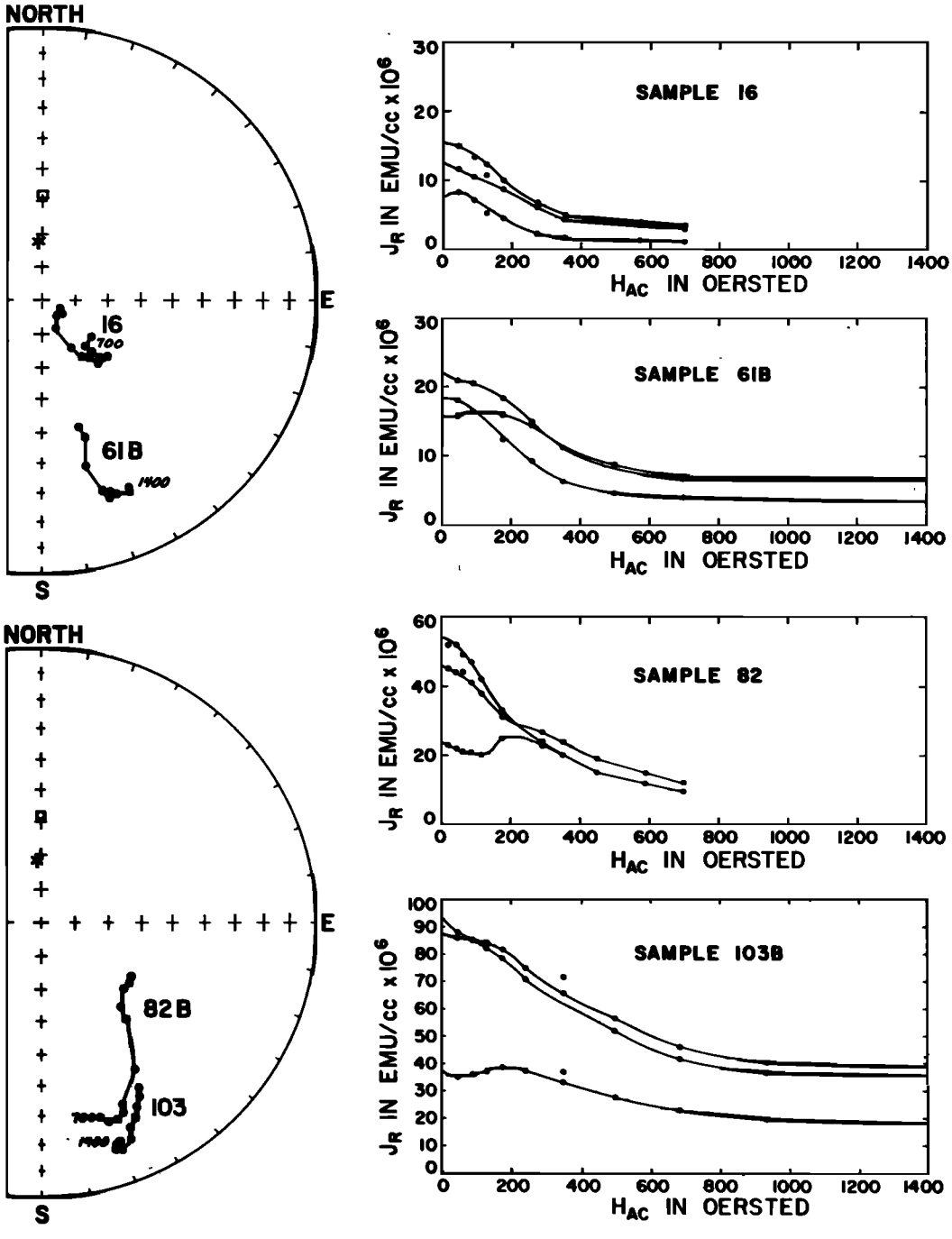


Fig. 4. Complete ac demagnetization curves for four of the samples studied.

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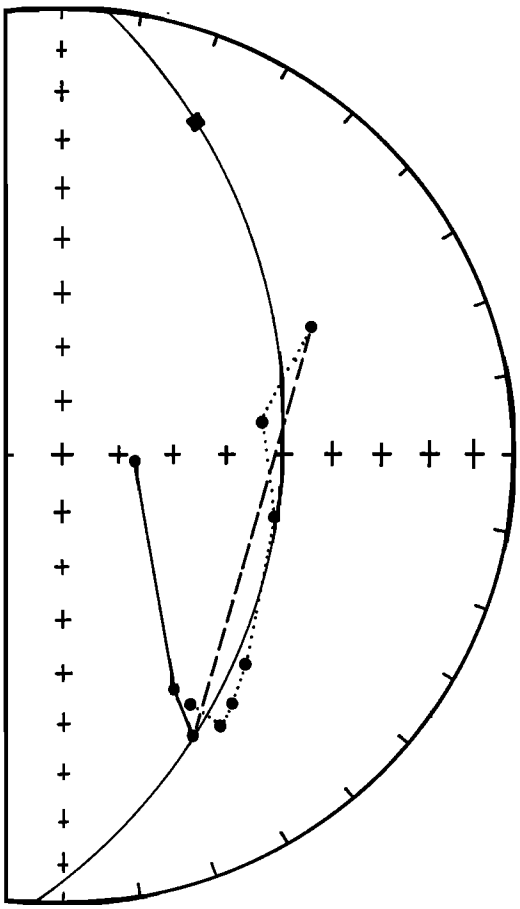


Fig. 5. Demagnetization experiments on sample 46*B*. ■ orientation of laboratory applied field (see text for further explanation).

ples had been sitting in the laboratory in more or less random orientation for some two to ten weeks. Those samples from site 4 (Figure 6*a*) show changes well in excess of instrumental errors for about half of the samples, but apparently no systematic change has occurred. However, the samples from sites 5 to 10 (Figure 6*b*) show large changes, and most of the change seems to have been in azimuth rather than inclination, suggesting that these samples might be able to acquire an isothermal component in the bedding plane more readily than at right angles to it. Further experiments are needed to see if this is indeed the case.

A comparison of the means for the original measurements and the remeasurements for sites 5 to 10 suggests that no significant change has taken place even though some of the individual measurements have changed by as much as 20°.

Since most of the change in direction upon demagnetization takes place at field intensities of less than 350 oe, a field intensity of 500 oe was chosen as a suitable value for treatment of large numbers of samples. Three sets of samples were treated in this manner. Figure 7*a* shows the results of the demagnetization of eleven of the thirteen samples from site 4. The complete demagnetization curve of one of these samples (61*B*) is shown in Figure 4. All of the samples moved away from the present field direction, but at least two and probably four of them did not move back into the cluster near the edge of the net. Instead, they lie on the instability great circle some 10° to 20° from the stable orientation. This suggests that some of the secondary component is 'hard' and cannot be preferentially removed by ac methods.

Figure 7*b*, showing the results of the demagnetization experiments on eleven of the samples from site 2, provides much the same con-

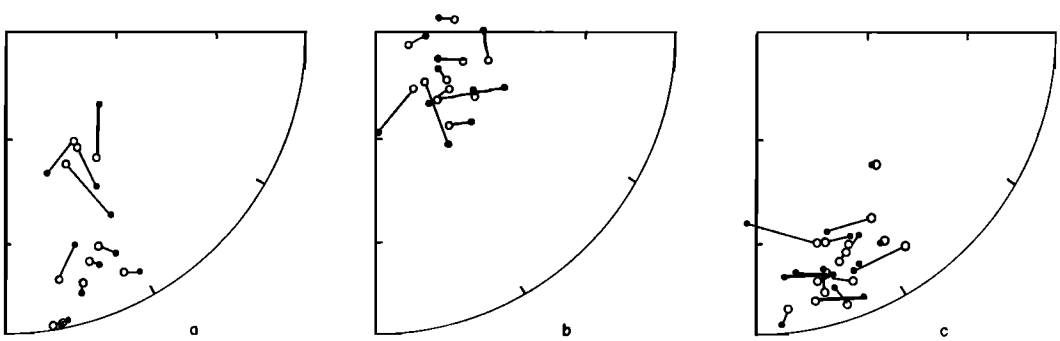


Fig. 6. Original measurements (·) and remeasurements (o): (*a*) site 4, (*b*) site 2, and (*c*) sites 5–10.

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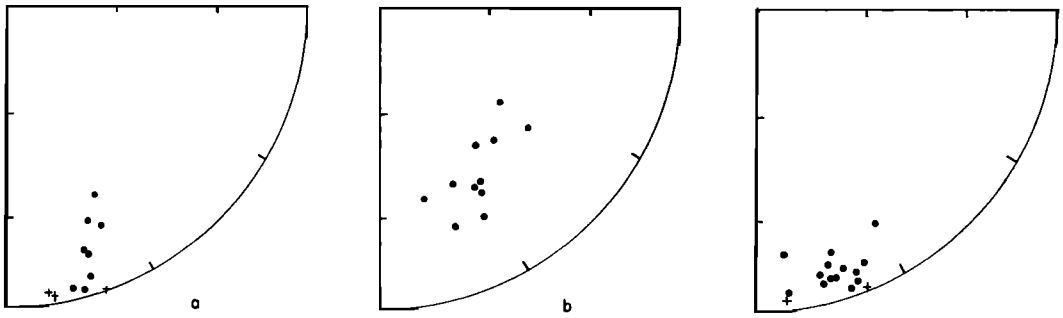


Fig. 7. Direction of remanent magnetization after ac demagnetization: (*a*) site 4, (*b*) site 2, and (*c*) sites 5–10.

clusion. The complete demagnetization curves for two of these samples (82*B* and 46*B*) are shown in Figures 4 and 5. All of these samples were certainly unstable, but their magnetizations do not move all the way back to the stable position given by the other localities. When these results are combined with those in Figure 7*a*, all the points lie near a great circle going through the Permian field direction (based on other localities) but not through the present field direction. Instead, they are apparently a sum of the Permian direction and a direction that can be attributed to a virtual magnetic pole of normal polarity, located somewhere between the Permian and present poles. It is possible that the samples from site 2 and site 4 may have acquired a 'hard' component at a later time in addition to the 'hard' original magnetization that is assumed to have been acquired during the Permian.

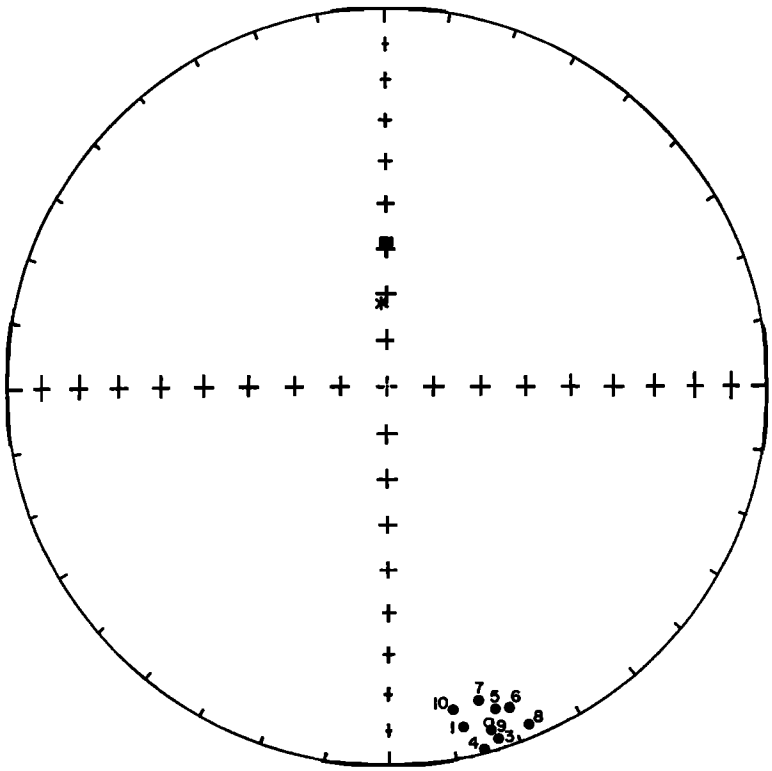


Fig. 8. Mean direction of magnetization for each site after ac demagnetization.

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Demagnetization of sixteen samples from sites 5 to 10 (Figure 7*c*) produced a more normal result, namely a decrease in over-all scatter and a change of about 8° in their mean directions of magnetization. The complete demagnetization curve for one of these samples (103*B*) is shown in Figure 4. The mean direction of the demagnetized samples for each of the six stratigraphic units is about the same as the mean direction for the high-density part of the cluster shown in Figure 3*e*.

The mean directions of magnetization for each site after demagnetization is shown in Figure 8. The changes from Figure 3*f* are not large but do produce a tighter grouping and thus imply a more reliable result. Table 1 provides a summary of these data and their calculated pole positions.

Comparison with other North American Permian results. The geomagnetic pole positions, assuming a geocentric dipole, resulting from this study are compared with those from other parts of North America and Europe in Figure 9. The great-circle routes connecting the source areas with the virtual pole positions (or mean position) are also shown. Because

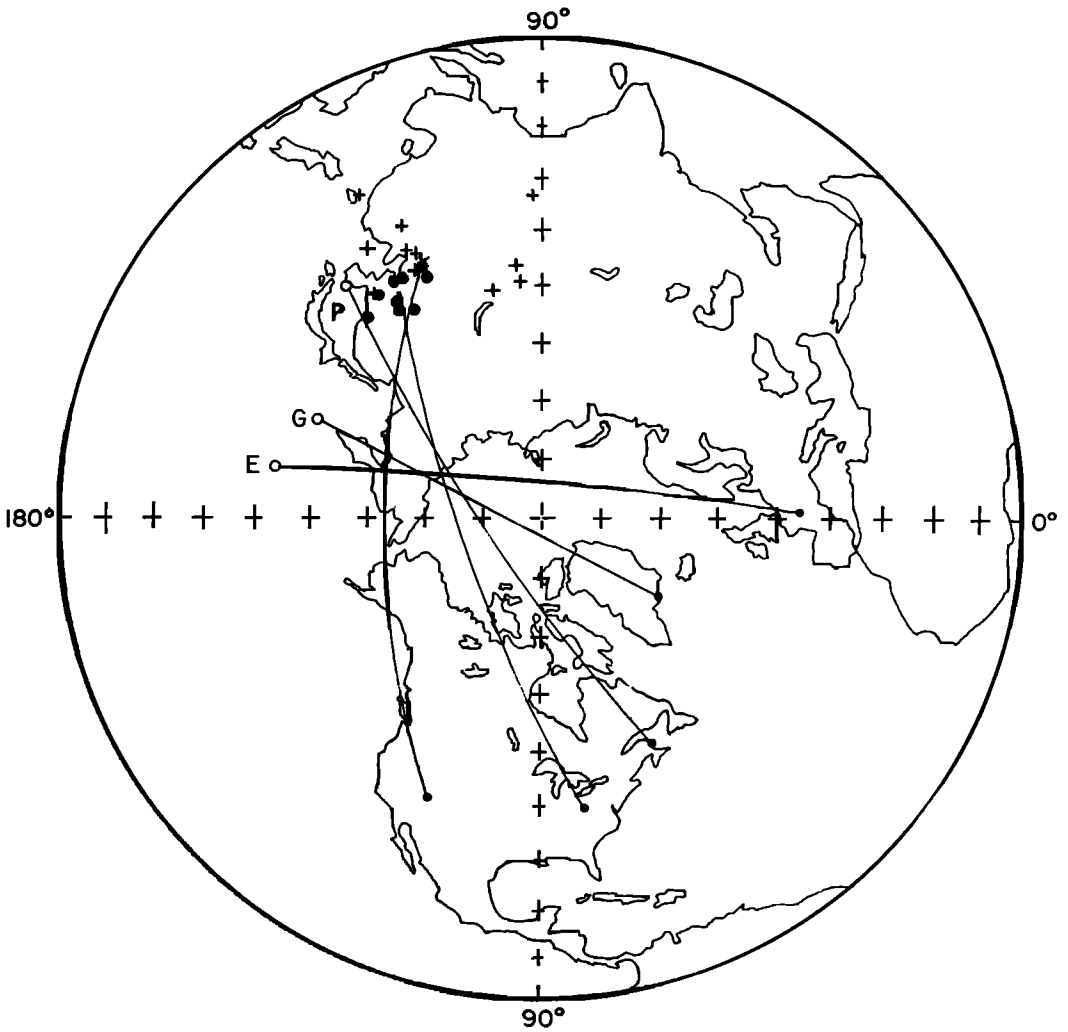


Fig. 9. Comparison with other Permian data from North America and Europe: ● Permian virtual poles from West Virginia, + Permian virtual poles from western United States, × mean of western United States, ○ mean Permian virtual poles for Europe (*E*) Greenland (*G*) and Prince Edward Island (*P*).

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different conclusions are reached in the two cases, the comparisons must be made in two parts, i.e., West Virginia versus western United States and West Virginia versus Prince Edward Island, Greenland, and Europe. First to be considered are the implications arising from the comparison with the data from the western United States.

The pole positions derived from Permian rocks in the western United States are shown by crosses in Figure 9. Their mean is given by the cross at the end of the great circle that joins it with their general source area in the southwestern United States. Unfortunately, this mean direction is subject to change, depending upon the weightings given the various results. The mean shown was derived by equally weighting all of the formations and areas given in *Cox and Doell* [1960]. This mean is not markedly different from that computed by *Nairn* [1963]. However, the grouping of virtual pole positions used for this mean is not good (about 30° spread) and is not significantly improved by limiting the control points to just those from the Supai formation [*Cox and Doell,* 1960, p. 747]. This spread which appears excessive when compared with that found in the Dunkard rocks (heavy dots) probably represents a period of time comparable with that represented by most of the Supai results but does not represent the same extent or geographic distribution. If the mean from the western United States is assumed to be correct, it is interesting to note that the two sets of results are not distinguishable in either azimuth or inclination, as shown by the intersection of the great circles at the mean positions for the two groups. This result supports the geocentric dipolar hypothesis for the past magnetic field in two ways. First, a test based on inclination alone gives the same paleolatitude, but this test could be in error because of inclination errors [*Cox and Doell,* 1960]. A second test, independent of inclination and thus free of possible inclination errors, can be made by noting the intersection point given by the paleodeclinations (as shown in Figure 9). Since this intersection provides the same distance to the pole as the inclination data, it suggests that the dipolar hypothesis is correct and that there is no inclination error associated with these sediments.

Since the above conclusion is dependent on the western mean direction being the true mean direction, it is interesting to note that when the West Virginia results are compared with the results from Prince Edward Island (*P*) [*Fuller,* 1963], Greenland (*G*) [*Bidgood and Harland,* 1961], and the European mean direction (*E*) [*Nairn,* 1963], all of the great circles cross in a small area in northern Siberia. This may imply that the field was nondipolar or that a large inclination error is involved in all samples. A third possibility may be that this intersection of the azimuth lines is an artifact of the tectonic rotation involved in the opening of the Atlantic Ocean basin during the westward drift of North America [*Carey,* 1958; *Bidgood and Harland,* 1961]. The choice from among these three possibilities is dependent upon re-evaluation of the scatter associated with the results from the western United States, additional information from eastern North America, and the Arctic, and a check for inclination error by measuring the NRM of associated red beds and lavas. Such a study is now in the preliminary stages.

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